

Choosing a database

If you have ever designed or worked on an application that stores data in a persistent manner then you are most likely familiar with a database of some kind. While choosing a database might seem trivial compared to designing the logic layer sometimes a choice in the database technology can actually affect your application logic.

For example: A company which had chosen to migrate to an Oracle database from a mysql one at some point of time, in order to comply with client guidelines, realised that some of their queries on production were not working on the new database! A quick investigation revealed that Oracle limits the number of positional parameters in a query to 2000, post which it throws an exception! Eventually, they had to rewrite the entire application logic around their new database system.

When choosing a database of some kind, be very clear about the kind of data that you are expecting to store in it now and in the near future. Whether it's relational or has large objects or needs very fast access time. Also, on the percentage of read or write operations, the choice of a database for specific uses (like archiving) might vary.

Choosing language(s) for development

- `python > java ? python : java;`

While the above statement might have drawn more than a few cringes from the readers it remains a valid subject, that once we have our technology stack in place we would need to decide on the language of our choice to code in.

Of course a qualitative analysis on the benefits of choosing a strongly typed or a weakly typed language, an object oriented or a procedural one or one that supports functional programming would be too long and likely cause a small debate war but what we can attempt to do is to ensure that we choose one that has a clear syntax, has good IDE support, a solid collection of libraries with good API support for most if not all of the technologies in our stack.

Another key factor here is the learning curve for the developers using the language, while some on-hand learning is expected, throwing a dev team into a war room with a completely new language and asking them to create a well formed application in a month would most likely not yield expected results.